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Enhancing methane production from dewatered waste activated sludge through alkaline and photocatalytic pretreatment

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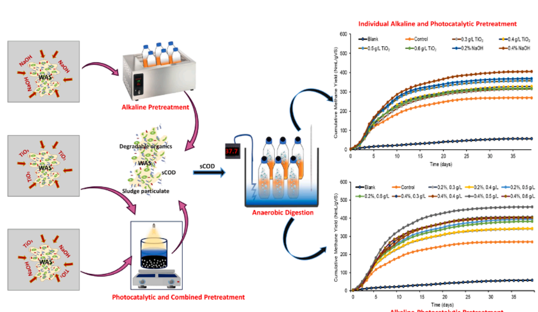
HIGHLIGHTS

- Alkaline, photocatalytic and combined pretreatment of dewatered sludge was studied.
- Individually, alkaline pretreatment disintegrated the sludge more than photocatalysis.
- Alkaline and combined pretreatment resulted in 11 and 37% disintegration degree.
- Methane yield enhanced by 50 and 71% due to alkaline and combined pretreatment.

GRAPHICAL ABSTRACT



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ABSTRACT

Waste activated sludge generated from wastewater treatment plants makes an abundant source of biomass. Its effective utilization through anaerobic digestion (AD) requires pretreatment to disintegrate the sludge matrix and increase organic matter availability. In this study, dewatered waste activated sludge (DWAS) was subjected to alkaline, photocatalytic, and alkaline-photocatalytic pretreatment for its disintegration and subsequent methane production using different concentrations of sodium hydroxide and titania nanoparticles. Individual pretreatment resulted in maximum disintegration degree (DD_{sCOD}) of 11.3 and 5.2% at 0.8% NaOH and 0.6 gTiO_2/L, respectively. Alkaline-photocatalytic pretreatment yielded 37% DD_{sCOD} at 0.8% NaOH–0.4 g/L TiO_2. As compared to control, AD at 0.4% NaOH and 0.5 g/L TiO_2 pretreatments yielded maximum methane, which was 50.4 and 32.6% higher. Similarly, alkaline-photocatalytic pretreatment at 0.4% NaOH-0.5 g/L TiO_2 yielded methane as 462 N mL/g VS, which was 71.1% higher. Modified Gompertz model fitted the methane yield data well.

1. Introduction

Huge amounts of sewage sludge and waste activated sludge (WAS) are being continuously produced as a leading by-product from different conventional and non-conventional wastewater treatment processes (Alqaralleh et al., 2019). The production rate is estimated to reach 60

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million tons of wet sludge annually that needs extensive treatment facilities for proper handling (Wacławek et al., 2019; Zhang et al., 2017). Sludge treatment and safe disposal burdens 50–60% of wastewater treatment plants' (WWTP) total operational cost (Zhen et al., 2017). If left untreated, certain harmful substances such as organic and inorganic contaminants, heavy metals, and pathogens can contaminate the environment and cause hygiene and public health concerns (Alqaralleh et al., 2019). WAS being a rich source of organic matter and nutrients may generate bioenergy and meet the rising energy demands. Among various treatment and conversion techniques, anaerobic digestion (AD) reduces the weight and volume of sludge and generates eco-friendly biogas (Zhang et al., 2017).

Although AD decreases the solid content of sludge, there is still a considerable gap to improve the hydrolysis of macro-organic substances, which is limited by the rigid structure of flocs and microbial cell wall. Therefore, pretreatment is required to maximize the utilization of sludge organic matter and improve rate-limiting hydrolysis and digestibility during AD. Pretreatment disintegrates sludge flocs and microbial cell wall, releasing internal organic materials of cells and causing relatively easier solubilization, thus enhancing methane production. Methane produced by AD of WAS can meet 70–80% of the energy consumed by WWTPs if the digestion efficiency is maximized (Wang and Li, 2016). Numerous sludge pretreatment technologies such as chemical, thermal, mechanical, ozonation, microwave irradiation, advanced oxidation process (AOP), ultrasonic and enzymatic methods, and their combinations have been studied to overcome the limitations (Kavitha et al., 2016; Ribeiro et al., 2015; Sharmila et al., 2019; Yuan et al., 2019a, 2019b).

Alkaline pretreatment is known to dissolve and destroy sludge flocs using hydroxyl ion by inducing swelling and releasing soluble organics (Zhen et al., 2014). A relatively low dose is effective in solubilizing the insoluble organic matter into soluble organic carbon forms. Studies unveiled an increase in methane production with a decrease in volatile solids at low-dose alkaline pretreatment (Li et al., 2012; Lin et al., 2009; Shao et al., 2012). Many studies preferred the use of NaOH to other alkalis, which causes relatively greater solubilization of organic matter. A study demonstrated solubilization of WAS at doses ranging from 20 to 80 mg NaOH/gTS, resulting in 10.36% increase in organic matter degradation rate at 80 mg NaOH/gTS and 34.68% biogas enhancement at 60 mg NaOH/gTS over control (Zhang et al., 2015).

Advanced oxidation process has gained significant importance for solubilizing and improving the hydrolysis rate in AD by degrading organic matter to soluble form, using semiconductor photocatalyst titanium dioxide (TiO_2) (Arlos et al., 2016). TiO_2 nanoparticles (NPs) have been widely used as an efficient and environmentally safe photocatalyst for the degradation of organic matter due to its chemical stability, low toxicity, low cost, and high photoactivity. When irradiated with ultraviolet (UV) light as a photon source, hydroxyl radicals are generated, which disintegrate the organic matter due to photo-induced oxidative reaction (Anjum et al., 2018b). A study demonstrates how TiO_2 NPs at 0.4 g/L TiO_2 increased soluble chemical oxygen demand (sCOD) up to 18.4% over control WAS, which generated 235.6 mL/gVS methane along with suspended solids reduction up to 53.8% (Sharmila et al., 2017).

Recently, combined pretreatment using different techniques has gained significance as an energy-efficient option as they solubilize sludge and obviate the need for higher pretreatment concentrations. Several research works have confirmed the synergistic effect of combined pretreatment on AD of sludge (Tian et al., 2018). It was found in a recent study that combining alkaline pretreatment and ultrasonication increased sCOD from 1200 to 18,000 mg/L (Tian et al., 2016). Another research conducted combined alkaline and microwave irradiation pretreatment on sludge at optimal pretreatment condition of 0.1 g CaO_2/g VS – microwave at 480 W and 2 min, resulted in 80.2% higher methane yield as compared to control (Wang and Li, 2016).

Yet, to the best of our knowledge, no study exists on assessing the effectiveness of combined alkaline-photocatalytic pretreatment on sludge solubilization and subsequent AD. However, a study on combined pretreatment of rice straw using 2 g/LTiO_2 NPs and 1.5 wt% NaOH led to an increase in rice straw cellulose from 37.5% to 71.5%, and decrease in lignin from 18.5% to 9.0% (Niu et al., 2009). In this study, two different types of pretreatments were applied to assess their individual and combined effect on characteristics and AD of WAS. It would reduce the required quantity of chemicals and energy applied and enhance the efficiency of WAS digestion.

2. Materials and methods

2.1. WAS and inoculum preparation

WAS was collected from the bio-tank of a full-scale aerobic MBR (Membrane Bioreactor) plant treating wastewater of NUST, Islamabad. The collected sludge was thickened, followed by dewatering the settled faction using a sand drying bed. The dewatered WAS (DWAS) was stored at 4 °C for further use and analysis. Inoculum consisted of a mixture of anaerobic sludge sediments collected from the constructed wetland facility of NUST, Islamabad, and digested cow manure collected from a digester in the vicinity. The inoculum was developed by incubating both the materials separately at 37 °C under anaerobic conditions for twenty days and then homogeneously mixing them at a ratio of 1:1 on total solids basis to enhance microbial diversity (Tian et al., 2018).

2.2. Synthesis and characterization of photocatalyst

Titanium (IV) Oxide Anatase grade, Daejung, was used as a precursor for the preparation of TiO_2 NPs. Nanoparticles were synthesized using liquid impregnation method. For this purpose, 300 mL solution of 167 g/L TiO_2 was stirred for 24 h, followed by 24 h settling. The obtained slurry was dried in an oven (Memmert, UN110, Germany) at 105 °C for 12 h, which was then crushed and calcined in a muffle furnace (Barkmeyer, NEY M-525 S II, U.S.A) at 550 °C for 6 h. The resultant powder was allowed to cool down at room temperature and then crushed to obtain clear crystalline form of titania nanoparticles (Husnain et al., 2016). Characteristics of TiO_2 NPs were analyzed through scanning electron microscopy (SEM) and energy dispersive spectroscopy (EDS) (Tescan, Vega 3, Czech Republic). SEM analysis confirmed the size of TiO_2 NPs to be in the range of 44–50 nm under operating conditions, i.e., voltage: 20 kV, magnification: 44.4 kx, working distance: 6.92 mm, and view field: 500 nm. EDS linked with SEM confirmed the purity of NPs, indicating O_2 59.01 and Ti 36.11%.

2.3. Alkaline pretreatment

Alkaline pretreatment was carried out using NaOH solutions of 0.2, 0.4, 0.6, 0.8% w/v concentrations. The NaOH solutions were prepared by adding the required NaOH pellets (Analytic-ACS, Germany) in distilled water, which was then mixed at 250 rpm until the solution became clear. The pretreatment of DWAS was performed at TS of 10% (dried solids to solution ratio 1:10), which gave the specific alkali doses of 2, 4, 6, and 8 g NaOH/100 g TS_{sludge}. The pretreatment at each concentration was carried out in 500 mL Pyrex glass bottles with 350 mL working volume. After adding the required quantity of NaOH solutions in the pre-weighed quantity of sludge solids, stirring was performed for 5 min at 200 rpm using a magnetic stirrer to ensure homogeneity of the suspension. Pretreatment was carried out for 3 h at 37 °C in a water bath. After pretreatment, each sample was neutralized to pH 7 by adding 0.02 N H_2SO_4 (Li et al., 2013).

2.4. Photocatalytic pretreatment

The photocatalytic pretreatment of DWAS was carried out in a 1L beaker with a working volume of 600 mL, placed in an enclosed chamber equipped with 4 UV-B lamps, each of 100 W power and 320 nm wavelength. Pretreatment was carried out at TS of 10%, under continuous stirring for 3 h at 37 °C and TiO_2 doses of 0.3, 0.4, 0.5, and 0.6 g/L.

2.5. Alkaline-photocatalytic pretreatment

For alkaline-photocatalytic pretreatment, each dose of NaOH used in alkaline pretreatment was combined with all TiO_2 NPs doses used in photocatalytic pretreatment. Pretreatment conditions were the same as described in Section 2.4. Each pretreatment was performed in triplicate, and all the pretreated samples were neutralized.

2.6. Batch anaerobic digestion

The anaerobic digestion experiment was performed using DWAS pretreated with optimized NaOH doses (0.2 and 0.4%), all doses of TiO_2 NPs (0.3, 0.4, 0.5, and 0.6 g/L TiO_2), and their combinations. The optimum NaOH doses were selected on the basis of characteristics of pretreated DWAS. For anaerobic digestion, serum bottles with a total volume of 300 mL, and a working volume of 225 mL were used as batch reactors. The substrate to inoculum ratio was kept 1:1 (on the basis of gVS), whereas substrate loading was kept 10 g VS/L. Blank reactors having inoculum were incubated in parallel to provide the background biogas production from inoculum alone. Therefore, sixteen sets of reactors were prepared in triplicate viz., inoculum, raw WAS, NaOH pretreated WAS, TiO_2 NPs pretreated WAS, and alkaline-photocatalytic pretreated WAS. The reactors were sealed with a rubber septum and crimped, then flushed with nitrogen gas to ensure anaerobic conditions and were placed in an incubator (VELP, FOC 120E, Italy) at 37 °C for 40 days. Manual shaking was provided to every reactor twice a day.

2.7. Analytical techniques

The characteristics of DWAS were analyzed before and after pretreatment to determine the effectiveness of pretreatment. Measurement of total solids (TS), volatile solids (VS), total chemical oxygen demand (tCOD), soluble chemical oxygen demand (sCOD), total Kjeldahl nitrogen (TKN), and total ammonia nitrogen (TAN) was performed by standard methods (APHA, 2017). pH was measured using a pH meter (HANNA, model 8521, U.S.A). Disintegration degree (DD_{sCOD}) was calculated using Eq. (1), which shows the degree of sludge particulate substances solubilized by pretreatment (Tian et al., 2018).

$$DD_{sCOD}\% = \frac{sCOD_o - sCOD_o}{tCOD_o - sCOD_o} \times 100 \quad (1)$$

where $sCOD_p$ is the sCOD of pretreated sludge, $sCOD_o$ is the sCOD of untreated sludge, and $tCOD_o$ is the tCOD of untreated sludge.

Organic matter removal efficiency (OMRE) was calculated based on sCOD, TS, and VS reductions from digesters after AD. SEM analysis of DWAS was performed to observe the change in surface morphology after pretreatment. For SEM analysis, pretreated DWAS samples were neutralized, centrifuged, and air-dried (Zhen et al., 2014). After drying, samples were crushed into powder and oven-dried at 105 °C to remove moisture.

To measure soluble substances, the supernatant of sludge samples was obtained after centrifugation at 10,000 rpm for 10 min, followed by filtration through a 0.45 μm filter (Lu et al., 2018). Volatile fatty acids (VFA) and total alkalinity (TA) for anaerobic digesters were measured through the titration method (APHA, 2017).

Daily biogas production was recorded using the water displacement method (Yuan et al., 2019a). The digestion test ended when no more significant biogas production was observed. The biogas samples' methane content was analyzed by gas chromatography (GC-2010 Plus, Shimadzu, Japan) and multiplied with biogas volume to calculate the volume of methane present in the biogas. The gas volume was converted to normalized volume (i.e., N mL) and presented as dry gas, according to Eq. (2) (Dinuccio et al., 2010):

$$V_N = (V \times 273 \times (760 - -p_w))/((273 + T) \times 760) \quad (2)$$

where V_N = volume of dry biogas at normal temperature and pressure in NmL, V = recorded volume of the biogas (mL), p_w = water vapor pressure as a function of ambient temperature (mm Hg), and T = ambient temperature (°C).

2.8. Data analysis

In this study, modified Gompertz model was used for kinetic model fitting of the observed methane production data by Eq. (3) (Yu et al., 2019):

$$M(t) = P\cdot exp\left\{ - exp\left[\frac{R_{max}\cdot e}{P_m}(\lambda - t) + 1\right]\right\}a \quad (3)$$

where M(t) is the cumulative methane yield (N mL CH_4/g VS) with respect to AD time t (d), R_{max} is the maximum methane production rate (N mL CH_4/gVS-d), P_m is the maximum methane potential (N mL CH_4/g VS) at the end of AD, λ is the lag phase time (d). The kinetic estimation was performed using the IBM SPSS Statistics 16.

All the tests were done in triplicate, and the results were expressed as mean; standard deviations were also determined. A one-way analysis of variance (ANOVA) was employed to analyze the statistical significance of methane and solids removal using the Origin Pro 9.0. The correlation was considered statistically significant at a confidence interval of 95% with $p < 0.05$ probability.

3. Results and discussion

3.1. Characteristics of substrate and inoculum

Initial characteristics of substrate and inoculum are shown in Table 1. Substrate's characteristics play an important role in the pretreatment and subsequent digestion process. Results show that the TS content of DWAS was 10.10%, which was 6.6 times that of raw WAS having 1.54% TS. The reason is that waste activated sludge was dewatered before use in this study. Similarly, total COD and soluble COD of the DWAS was 29.9 and 0.196 g/L, respectively. The total COD of dewatered sludge varies depending on the level of dewatering. Total and soluble COD of sludge ranges typically from 10 to 40 g/L and 0.16 to 1.7 g/L, respectively (Yu et al., 2019; Yuan et al., 2019a, 2019b; Koupaie et al., 2017; Zhen et al., 2014).

3.2. Effect of alkaline and photocatalytic pretreatment on characteristics of DWAS

3.2.1. COD solubilization and sCOD release from DWAS

Alkaline pretreatment with NaOH increased sCOD concentration with the increase in NaOH dose. sCOD is regarded as the main parameter for evaluating the bioavailability and biodegradability of the DWAS particulate matter. An increase in sCOD may increase biogas production during AD due to increase in soluble organic matter. Fig. 1a shows the

Table 1
Characteristics of dewatered waste activated sludge and inoculum.

Parameters	Unit	DWAS	Inoculum
TS	%	10.07 ± 0.13	12.5 ± 0.19
VS	%TS	77.30 ± 0.32	57.9 ± 0.53
pH	–	7.3 ± 0.1	8.2 ± 0.2
Total COD	g/L	29.9 ± 0.18	21.2 ± 0.37
Soluble COD	g/L	0.196 ± 0.004	0.622 ± 0.02
TKN	%	3.4 ± 0.15	4.1 ± 0.6
TAN	mg/kg	428 ± 9	237 ± 2.8

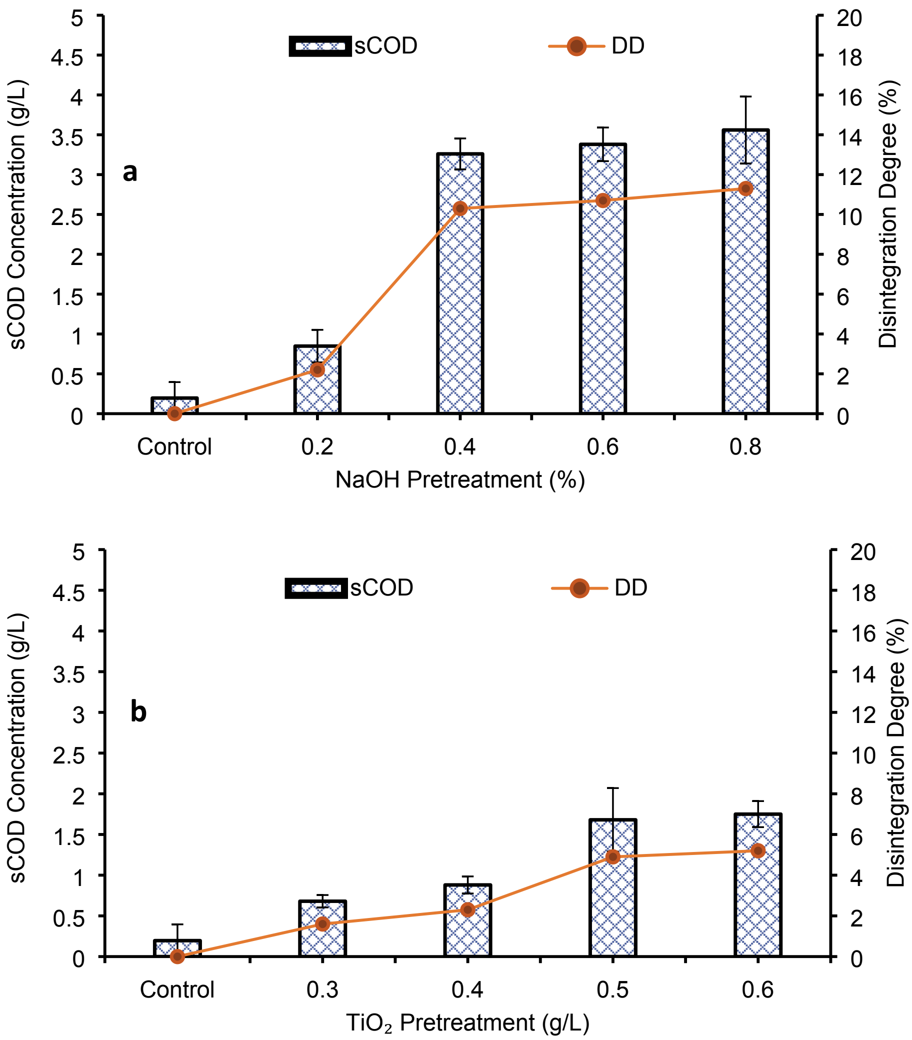


Fig. 1. Effect of alkaline (a) and photocatalytic (b) pretreatment on soluble chemical oxygen demand (sCOD) and disintegration degree (DD) of dewatered waste activated sludge.

effect of alkaline pretreatment on sCOD and DD_{sCOD}. It is evident from the figure that the sCOD at 0.4% NaOH pretreated DWAS was 4.7 times more solubilized than that of 0.2% NaOH pretreated DWAS as depicted by DD_{sCOD}. However, the increase in sCOD was not that remarkable with further increase in NaOH dose beyond 0.4% NaOH as the DD_{sCOD} value increased by 3.88 and 9.71% at 0.6 and 0.8% NaOH, respectively, as compared to 0.4% NaOH dose. Alkaline pretreatment creates an environment where microbial cells lose their integrity, resulting in improved sludge hydrolysis rate by transforming insoluble organic compounds into soluble compounds. These observations correlated well with those of Zhang et al. (2015), who observed 7.28 times higher sCOD than control at 0.4% NaOH but the sCOD increased at a decreasing rate beyond this dose. Similarly, Li et al. (2013) observed that the process of sCOD disintegration slowed by increasing the substrate's pH using 0.1 mol/L NaOH. DD_{sCOD} improved upon pretreatment due to conversion of COD into soluble fractions and profoundly increased by increasing alkali doses up to 0.4% NaOH, beyond which the increase was not noteworthy. The low rate increase in DD_{sCOD} beyond 0.4% NaOH in the current study demonstrates limited enhancement of sludge hydrolysis at 0.6 and 0.8% NaOH, and the findings of this study are well in line with those of Zhang et al. (2015).

Photocatalytic pretreatment with TiO_2 NPs increased the soluble organics of DWAS. Pretreatment showed an increasing trend in the concentration of sCOD with the increase in TiO_2 NPs dose, as shown in Fig. 1b. Results indicate that increase in COD solubility was observed with increase in TiO_2 NPs dose but up to a 0.5 g/L dose, at which DD_{sCOD} was 4.9%. Increase in DD_{sCOD} with the increase in TiO_2 NPs dose shows that the disruption of microbial flocs in DWAS might have occurred, resulting in the release of intracellular and extracellular substances. At 0.6 g/L TiO_2 NPs dose, DD_{sCOD} was just 5% higher than that at 0.5 g/L. Likewise, the sCOD concentration increased greatly at 0.4 and 0.5 g/L TiO_2 NPs dose, but at 0.6 g/L, the rate of increase slowed down. The decrease in COD's solubilization rate might be due to immediate mineralization of intracellular substances by the generated hydroxyl radicals, which are highly oxidative in nature, causing loss of organics from the substrate. Sharmila et al. (2015) reported 18.5% COD solubilization after 42 h pretreatment at an optimized TiO_2 dosage of 0.03 g/g SS and solar radiation with an average UV light intensity (5.52–6.06 kW h/m^2 day). DD_{sCOD} observed in the present study for photocatalytic pretreatment was lower than that reported by Sharmila et al. (2015). This might be due to high UV intensity, TiO_2 NPs dose, duration, and low sludge concentration used in their study compared to the present study.

3.2.2. Volatile solids reduction of DWAS

Alkaline pretreatment of DWAS caused a decrease in VS content from 77.3% to 68.9%, increasing alkali doses (0.2–0.8% NaOH). However, the reduction was not substantial as compared to raw DWAS, showing negligible removal efficiencies. Lin et al. (2009) reported 6% to 19% decrease in VS during alkaline pretreatment due to progressive hydrolysis of complex organic matter. At 0.8% NaOH dose, a maximum decrease of 8.4% in VS was observed in the present study, which is well in line with the findings of Lin et al. (2009).

The VS content of DWAS decreased from 77.3% to 71.2% with an increase in TiO_2 NPs doses (0.3–0.6 g/L TiO_2), resulting in only 7.9% organic matter degradation due to photocatalysis. The slight decrease in VS was associated with partial degradation, confirming that most of the solubilized DWAS remained in the system in soluble form. This overall removal of sludge organic matter during TiO_2 NPs pretreatment was due to the generation of active radicals (Ribeiro et al., 2015). Anjum et al. (2018b) reported 10.44% (92.1%–83.5%) decrease in VS after 6 h photocatalysis using 0.5 g/L TiO_2 NTs. Less reduction of VS in the present study is attributed to comparatively higher amount of organic matter in the sludge solution.

3.2.3. Total ammonia nitrogen of DWAS

Total ammonia nitrogen (TAN) is considered an important parameter as it can inhibit AD. TAN in DWAS increased gradually with an increase in NaOH dose. Concentration of TAN was observed to be 428.2, 498.2, 712.6, 655.2 and 532.6 mg/kg in DWAS after pretreatment at control, 0.2, 0.4, 0.6 and 0.8% NaOH respectively. Highest TAN at 0.4% NaOH resulted in 40% increase over control, whereas beyond 0.4%, a small decrease in TAN was observed at 0.6% NaOH, which further reduced at 0.8% NaOH. At higher doses, a gradual decrease occurred, which might be due to the volatilization of ammonia nitrogen. Pretreatment helps in the breakdown of inorganic matter and sludge nitrogen to ammonia nitrogen. Zhang et al. (2015) reported 50% increase in TAN at 0.6% NaOH pretreatment over control and a gradual decrease beyond this dose, which is well in line with the findings of the present study, indicating volatilization of ammonia nitrogen at higher doses of NaOH.

After photocatalytic pretreatment, TAN was 428.2, 478.5, 646.9, 696.4 and 619.2 mg/kg at control, 0.3, 0.4, 0.5 and 0.6 g/L TiO_2. Pretreatment at 0.5 g/L TiO_2 NPs resulted in the highest TAN value, which was 38.5% higher than control. Whereas it was 31% higher than control at 0.6 g/L TiO_2. Reduction in TAN concentration might be due to liberation of ammonia nitrogen at the highest pretreatment concentration.

3.3. Effects of alkaline-photocatalytic pretreatment on characteristics of DWAS

3.3.1. COD solubilization and sCOD release

The solubilization of sludge COD and its disintegration increased with an increase in alkaline-photocatalytic pretreatment doses, as shown in Fig. 2a. A considerable increase in sCOD was observed for the combinations of 0.4% NaOH with different TiO_2 doses. Similarly, combinations of 0.6% NaOH with TiO_2 NPs doses further increased sCOD availability by yielding maximum DD_{sCOD} of 30.5% at 0.6% NaOH-0.5 mg/L TiO_2, indicating higher availability of biodegradable matter as compared to control. Likewise, the combinations of 0.8% NaOH with TiO_2 doses further increased DD_{sCOD} over control. Maximum disintegration was observed at alkaline-photocatalytic pretreatment dose of 0.8% NaOH-0.4 g/L TiO_2, yielding DD_{sCOD} of 37%. This value of DD_{sCOD}

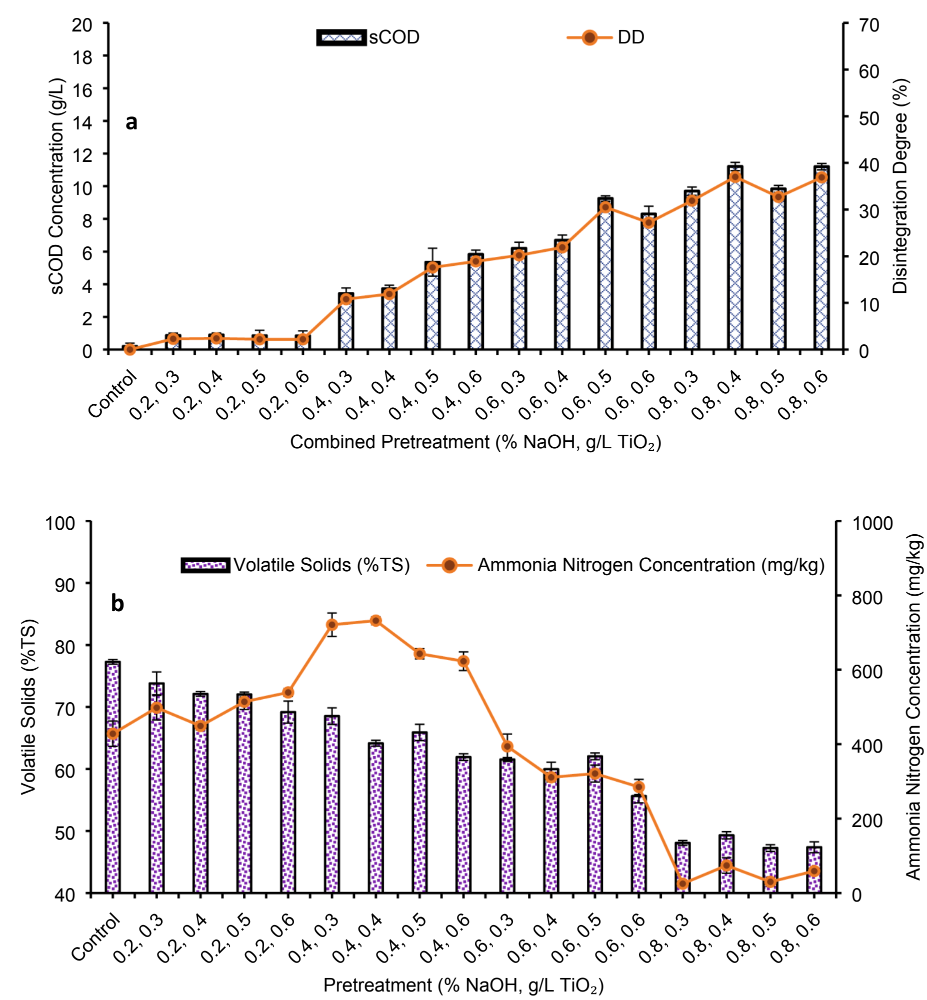


Fig. 2. Effect of alkaline-photocatalytic pretreatment on soluble chemical oxygen demand (sCOD) and disintegration degree (DD) (a), volatile solids and ammonia nitrogen (b) of DWAS.

is 69.5 and 93.8% higher than at 0.8% NaOH and 0.4 g/L TiO_2 NPs individual pretreatments, respectively (Fig. 1a and b). A considerable disintegration of sludge was observed during combined pretreatment using 0.4% NaOH and TiO_2 NPs, generating sCOD of 5.83 g/L. The transfer of COD to soluble fractions was greatly improved at higher NaOH doses (i.e., 0.6 and 0.8% NaOH) under various combinations with TiO_2 NPs. TiO_2 NPs and NaOH's combined influence resulted in higher solubilization of sludge organic matter. The remarkable rise in sCOD indicates the synergistic effect of alkaline-photocatalytic pretreatment on sludge organic matter disintegration. The addition of NaOH disrupts the extracellular biopolymeric substances surrounding the microbial cells making them more vulnerable to the effect of TiO_2 NPs. The reactive oxygen and hydroxyl radicals formed under UV radiations easily oxidize and convert organic matter of sludge into soluble form, resulting in higher cell lysis. The unprotected cells could not resist the harsh conditions induced by the combined effect and were effectively broken down into smaller units. Zhen et al. (2014) reported 54.7% DD_{sCOD} at 20 V and pH 12.2 through combined electrical-alkali pretreatment, indicating a synergistic disintegration effect. Although alkaline-photocatalytic pretreatment increased DDsCOD to 37% in the present study, it was lower than that of Zhen et al. (2014), probably due to the differences in the total solids and other properties of sludge being pretreated.

3.3.2. Volatile solids reduction

The effect of alkaline-photocatalytic pretreatment on VS content of DWAS indicates a reduction in VS with increased doses, as shown in Fig. 2b. The VS content varied between 73.8 and 47.38%, with a maximum decrease in VS content at alkaline-photocatalytic pretreatment dose of 0.8% NaOH-0.5 g/L TiO_2 NPs. This decrease was 31.4 and 32.6% higher than at 0.8% NaOH and 0.5 g/L TiO_2 NPs individual pretreatments (Section 3.2.2). The decrease in VS during combined pretreatment increased with high alkali loadings, contrary to individual pretreatments. These results indicate that alkaline-photocatalytic pretreatment removed certain fractions of organic matter from DWAS. The overall extraction of organic matter in alkaline-photocatalysis might be due to the generation of strong and active oxidizing agents i.e., $^{\bullet}OH$ and $O_2^{\bullet-}$, and swelling induced by NaOH, which increased the area of contact for degradation.

3.3.3. Total ammonia nitrogen

Effect of alkaline-photocatalytic pretreatment on DWAS initially increased TAN up to a certain level beyond which it decreased with an increase in pretreatment doses as illustrated in Fig. 2b. Enhanced effect of TiO_2 NPs in NaOH caused organic nitrogen, nitric organics such as nucleic acid, and proteins to convert into ammonia nitrogen, resulting in a 71% increase at 0.4% NaOH-0.4 g/L TiO_2 over control. This increase in TAN is 2.7 and 13.1% higher than that at 0.4% NaOH and 0.4 g/L TiO_2 NPs individual pretreatment (Section 3.2.3). Increased hydrolysis of proteins might have occurred until 0.4% NaOH. The gradual decrease in ammonia at 0.6% NaOH and higher doses might have been because of ammonia volatilization due to an increase in pH. At 0.8% NaOH, the synergistic effect of NPs showed a tremendous decrease in TAN to a minimum of 25 ± 8.8 mg/kg at 0.8% NaOH-0.3 g/L TiO_2. Alkaline-photocatalytic pretreatment might have caused hydrolysis of nitrogenous compounds and liberation of ammonia at high doses of NaOH. Li et al. (2017) reported an increase in ammonia nitrogen concentration by 79% over control at pH 9.0, 160 °C during alkaline-hydrothermal pretreatment, which agrees with the findings of the present study.

3.4. Effect of pretreatment on DWAS structure

Comparison of SEM images of alkaline-photocatalysis pretreated DWAS and raw DWAS showed disrupted floc structure of pretreated sludge. It showed conversion of macromolecules to micromolecules having compact structures with little void ratios. Exposure of sludge flocs to OH^- ions and $^{\bullet}OH$ radicals might have liberated entrapped organic matter. When exposed to alkali, the sludge flocs could not sustain turgor pressure and resulted in hydrolysis of the cell membrane and RNA in the cell and completely disrupted (Zhen et al., 2014). Zhu et al. (2019) suggested that in the course of photocatalysis, the pretreatment could denature the microbial cells and disabled the intracellular defense system and thus completely ruptured sludge flocs structure. Tyagi and Lo (2012) and Lin et al. (2009) made similar observations, reporting disrupted, and solubilized structure due to the enhanced pretreatment effect.

3.5. Effect of alkaline and photocatalytic pretreatment on anaerobic digestion of DWAS

The quality of pretreated sludge after alkaline and alkaline-photocatalytic pretreatment enhanced in terms of soluble COD as 11.3 and 37% of DD_{sCOD} was achieved at the highest doses, which provided the main indicator of pretreatment effectiveness. Individual photocatalysis did not show promising results after pretreatment, as presented in Section 3.2.1. It is noteworthy that no effective increase in sCOD production was witnessed by alkaline pretreatment beyond the dose of 0.4% NaOH. Similarly, there was an increase in DD_{sCOD} at 0.6 and 0.8% NaOH pretreatment but at a decreasing rate. Hence, the alkaline pretreatment doses of 0.2 and 0.4% NaOH were considered for the anaerobic digestion phase in individual setting and in combination with TiO_2 NPs doses.

3.5.1. Methane production from DWAS

The daily methane yield for DWAS after individual and combined pretreatment is presented in Fig. 3a and b, respectively. Highest daily methane yield of 52 N mL/g VS was observed for 0.4% NaOH followed by 0.2% NaOH and 0.5 g/L TiO_2 pretreated sludge with daily methane yield of 46 and 45 N mL/g VS, respectively. Control exhibited the lowest daily methane yield of 36 N mL/g VS. In the case of alkaline-photocatalytic pretreatment, 0.4% NaOH-0.5 g/L TiO_2 showed the highest daily methane yield of 51 N mL/g VS. Rapid generation of methane was noticed during the initial days of AD, which might be the maximum utilization of soluble substrate caused by dissolved organic matter in DWAS. Yu et al. (2019) reported a comparable shortened lag phase during digestion of WAS after pretreatment with choline supplement. In all individually pretreated samples, methane content remained stable. It varied between 55 and 65% after day 5 of digestion, whereas for control, it ranged between 52 and 60%. Zhang et al. (2015) reported that methane content was not significantly affected by alkaline pretreatment of sludge. Cumulative methane yield for all types of pretreatment is shown in Fig. 3c and d. Control sample exhibited the lowest cumulative methane yield of 270 N mL/g VS. In the case of alkaline pretreatment, 0.2 and 0.4% NaOH pretreated DWAS showed an enhancement of 36.7 and 50.4%, respectively, compared to control. Zhang et al. (2015) reported 35% increase in methane yield over control at 0.2% w/v NaOH pretreatment of dewatered activated sludge, which is in line with this study's findings. Studies on alkaline pretreatment of WAS showed progressive change in methane production and cumulative yield (Li et al., 2012; Wang and Li, 2016; Yuan et al., 2019a; Zhang et al., 2015). In the present study, 0.4% NaOH dose exhibited a difference of 13.7% from 0.2% NaOH due to sludge hydrolysis, which was sufficient to accelerate methanogenic activity. An increase in methane yield from alkali pretreated DWAS was significant for both the doses over control ($p < 0.05$). Moreover, there was a positive correlation ($r = 0.83$) between sCOD of alkali pretreated DWAS and its methane yield. Therefore, methane production was directly related to soluble form of COD produced due to alkaline pretreatment.

Photocatalysis pretreated DAWS at 0.3, 0.4, 0.5, and 0.6 g/L TiO_2 produced 315, 328, 358, and 318 N mL/g VS cumulative methane yield, showing an enhancement of 16.7, 21.5, 32.6, and 17.8%, respectively over control. Maximum methane production exhibited by 0.5 g/L TiO_2

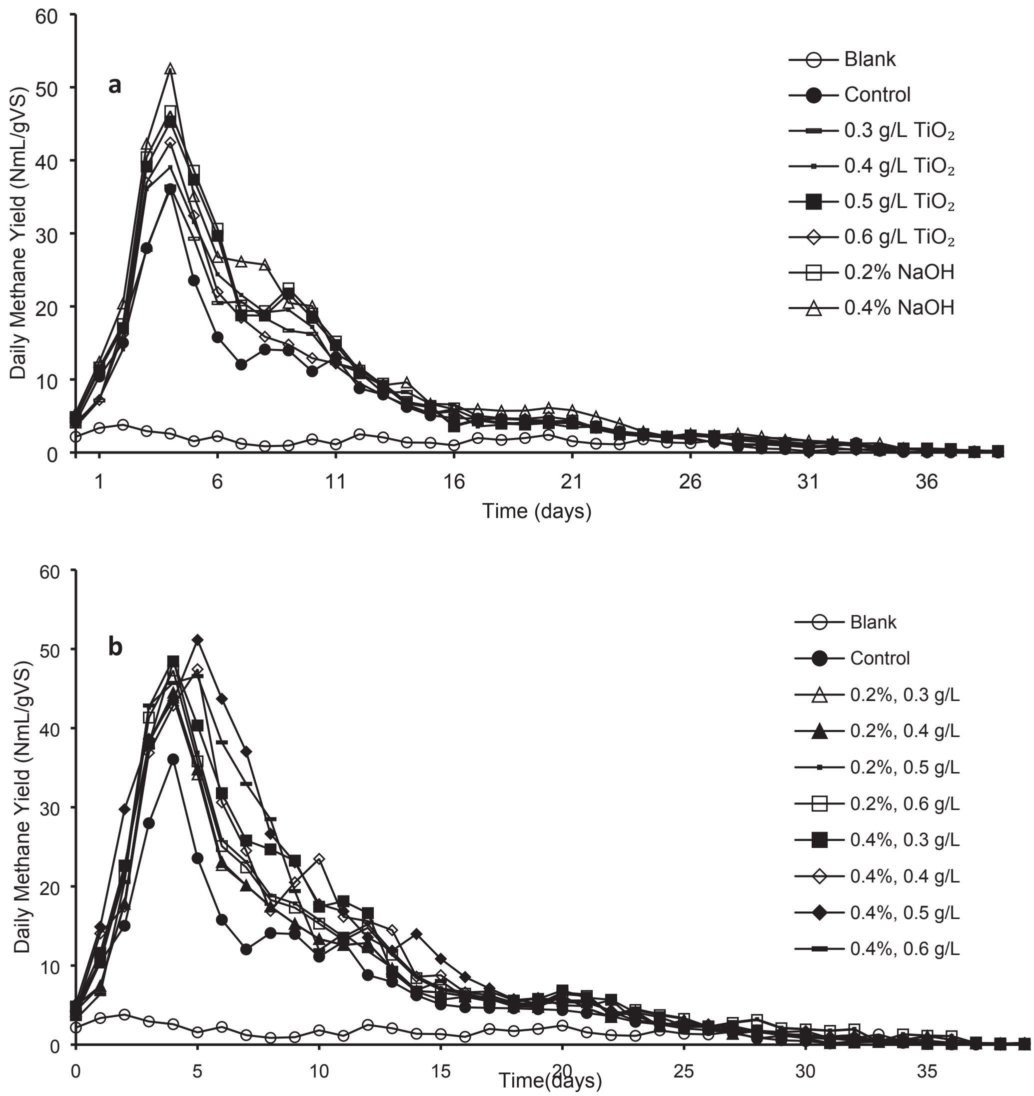
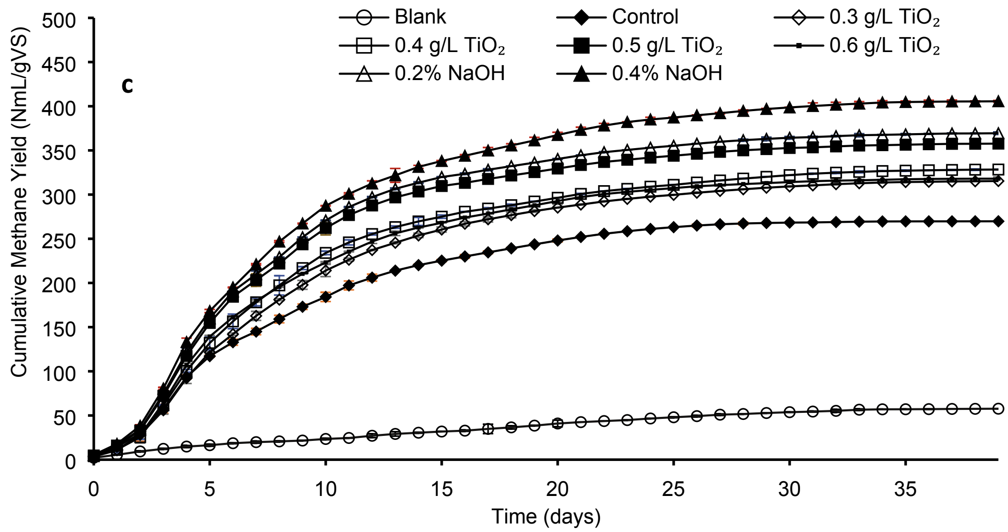


Fig. 3. Daily methane yield from NaOH and TiO_2 pretreated (a) and NaOH–TiO_2 pretreated dewatered waste activated sludge (b); Cumulative methane yield from NaOH and TiO_2 pretreated (c) and NaOH–TiO_2 pretreated dewatered waste activated sludge (d).

pretreated DWAS could be attributed to an increased sCOD value (Fig. 1b) as a positive correlation ($r = 0.77$) was noted between sCOD and methane yield of titania pretreated sludge. Anjum et al. (2018b) reported 37.7% increase in the yield due to photocatalytic pretreatment at 0.5 g/L TiO_2 NPs. Alvarado-Morales et al. (2017) also observed 37% increase in methane yield from the degradation of wheat straw by applying photocatalytic pretreatment using 1.5% w/w TiO_2 for 3 h. At 0.5 g/L TiO_2, 32.5% increase in methane yield was significantly different ($p < 0.05$) as compared to control but at 0.3, 0.4 and 0.6 g/L TiO_2 pretreatment, it was not significantly different ($p > 0.05$). The insignificant increase at 0.6 g/L might be due to low degradation at higher titania concentration, possibly caused by the increase in optical density of water, which is linked to increased viscosity, thus resulting reduction in penetration of UV light for photo-excitation of electrons. Other studies have also shown an overall positive impact of TiO_2 NPs on the degradation of WAS to improve sludge hydrolysis and methane yield (Godvin Sharmila et al., 2018).

Cumulative methane yield showed that alkali pretreated DWAS produced more methane than TiO_2 NP pretreated DWAS (Fig. 3c). Doses of 0.2 and 0.4% NaOH showed 36.7 and 50.4% enhanced methane, while the enhancement was 32.6% in the case of 0.5 g/L TiO_2 pretreated DWAS. This difference in production rate could be well explained through the contact area and the pretreating reagent's surface activity. TiO_2 NPs do not get dissolved and their activity is limited to contact at the surface (Anjum et al., 2018b). However, NaOH effectively accelerates swelling of particulate organics rendering destruction of polymeric sludge network, and subsequent exposure of entrapped cells leads to extra functioning of NaOH (Zhen et al., 2014).

Alkaline-photocatalytic pretreatment of DWAS showed further increase in cumulative methane yield as compared to control and individual pretreatments. 0.4% NaOH-0.5 g/L TiO_2 showed the highest methane production of 462 N mL/g VS, whereas 0.2% NaOH-0.3 g/L TiO_2 pretreated DWAS yielded the lowest cumulative methane as 341 N mL/g VS. Cumulative methane production for 0.4% NaOH-0.5 g/L TiO_2 was 71.1, 13.8 and 29.1% higher than control, 0.4% NaOH and 0.5 g/L TiO_2 individual pretreatments, respectively. Increased DD_{sCOD} of 17.6% at this pretreatment (Fig. 2a) depicts the availability of more soluble organic matter than control, which explains 71.1% enhancement in methane production. Increased methane production can be explained by disruption of extracellular biopolymeric substances surrounding the microbial cells upon the addition of NaOH, which makes them vulnerable to the effect of TiO_2 NPs. This might have disrupted sludge flocs,



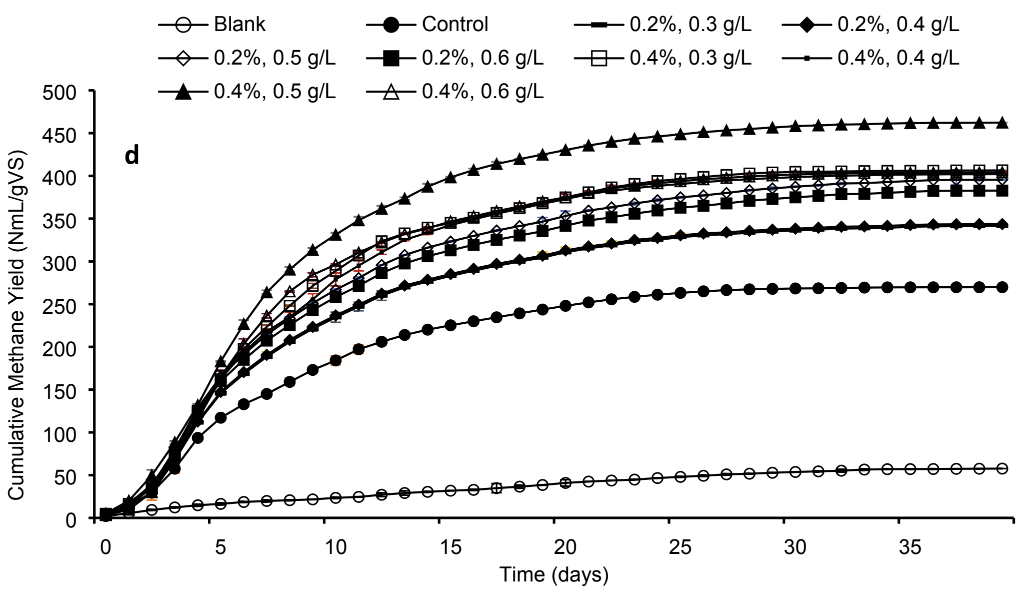


Fig. 3. (*continued*).

Table 2
Reactor stability parameters as affected by different pretreatments.

Anaerobic Digester	pH		VFA (mg/L)		TA (mg/L as $CaCO_3$)		VFA/TA	
	Initial	Final	Initial	Final	Initial	Final	Initial	Final
Blank	7.3 ± 0.1	7.2 ± 0.6	92 ± 7.1	131 ± 3.1	332 ± 6.5	632 ± 14.4	0.27	0.21
Control	7.3 ± 0.1	7.1 ± 0.3	131 ± 6.4	211 ± 2.2	389 ± 5.4	839 ± 13.1	0.33	0.25
0.3 g/L TiO_2	7.3 ± 0.3	7.1 ± 0.1	140 ± 3.4	198 ± 2.7	789 ± 6.1	879 ± 14	0.27	0.22
0.4 g/L TiO_2	7.3 ± 0.1	7.2 ± 0.6	184 ± 9.1	153 ± 2.8	641 ± 4.8	461 ± 8.4	0.28	0.33
0.5 g/L TiO_2	7.3 ± 0.1	7.2 ± 0.7	223 ± 7.1	229 ± 4.2	745 ± 9.5	845 ± 14.3	0.29	0.24
0.6 g/L TiO_2	7.3 ± 0.4	7.0 ± 0.5	201 ± 5.7	214 ± 3.5	738 ± 7.3	939 ± 11.7	0.27	0.22
0.2% NaOH	7.3 ± 0.3	7.0 ± 0.1	372 ± 8.4	472 ± 7.1	1430 ± 14.1	1730 ± 14	0.26	0.27
0.4% NaOH	7.3 ± 0.1	6.9 ± 0.4	487 ± 4.1	651 ± 11.2	1880 ± 17.4	1980 ± 18	0.25	0.21
0.2% NaOH-0.3 g/L TiO_2	7.3 ± 0.2	7.2 ± 0.6	377 ± 7.1	497 ± 2.6	1499 ± 18	1989 ± 16	0.25	0.24
0.2% NaOH-0.4 g/L TiO_2	7.3 ± 0.1	7.1 ± 0.4	392 ± 8.2	475 ± 3.7	1580 ± 18.6	1980 ± 18	0.24	0.24
0.2% NaOH-0.5 g/L TiO_2	7.3 ± 0.4	7.2 ± 0.1	402 ± 10	443 ± 7.9	1730 ± 14	1930 ± 11.3	0.23	0.22
0.2% NaOH-0.6 g/L TiO_2	7.3 ± 0.2	7.0 ± 0.3	417 ± 11.3	457 ± 7.2	1780 ± 17	1991 ± 7.1	0.23	0.21
0.4% NaOH-0.3 g/L TiO_2	7.3 ± 0.8	7.0 ± 0.2	559 ± 17.3	801 ± 17.3	2398 ± 25	2898 ± 19	0.23	0.18
0.4% NaOH-0.4 g/L TiO_2	7.3 ± 0.1	7.1 ± 0.8	601 ± 14.3	604 ± 13.7	2302 ± 21	2902 ± 11.8	0.26	0.20
0.4% NaOH-0.5 g/L TiO_2	7.3 ± 0.1	7.2 ± 0.4	687 ± 18.1	428 ± 8.1	2684 ± 29.1	2984 ± 18	0.25	0.14
0.4% NaOH-0.6 g/L TiO_2	7.3 ± 0.5	7.2 ± 0.1	524 ± 17.2	471 ± 10.7	2138 ± 26.2	1888 ± 11.3	0.24	0.20

increased surface area available for enzymatic action, and enhanced available organic matter. Methane production from alkaline-photocatalytic pretreatment of DWAS at all doses was found to be statistically different ($p < 0.05$) from control. In the past, the combined effect of alkali and TiO_2 has not been investigated for methane production from DWAS. However, Park et al. (2012) reported higher methane production as 404 mL/g VS after alkaline-ultrasound pretreatment of thickened pulp mill WAS. Wang and Li (2016) reported 25.4% methane enhancement relative to control due to CaO_2/microwave combined pretreatment. There was a positive correlation ($r = 0.75$) between soluble COD and methane yield of DWAS after alkaline-photocatalytic pretreatment. Therefore, methane production was directly related to soluble form of COD produced due to combined pretreatment.

3.5.2. Process stability of anaerobic digesters

The stability of batch anaerobic digesters was evaluated using operating parameters such as pH, VFA, TA, and VFA/TA. Deviation from optimum values in these parameters leads to acid accumulation, less methane generation, and anaerobic digester failure. Table 2 shows the initial and final values for these stability parameters. All the digesters were buffered externally to maintain pH in the range of 7.2–7.3, and this range lied within optimum pH values (6.8–7.4) needed for efficient methanogenesis in AD (Wang and Li, 2016). No noticeable decrease in pH was observed after 40 d digestion process and all the reactors had pH in the range of 6.9–7.2. pH value higher or lower than 8.5 or 6.5, respectively is not suitable for the proper functioning of methanogens.

Complex organic molecules in DWAS after pretreatment are converted to simpler forms that are accessible for acidogens to generate volatile fatty acids (VFAs), which are consumed in methanogenesis. VFA concentrations in the digestate of the reactors yielding higher methane decreased after digestion. Final values of VFAs in all the reactors of the present study were within the recommended limit for a healthy digestion system i.e., <1500 mg/L (Li et al., 2019), indicating stable consumption of acids produced during the digestion process by methanogens. Post-digestion total alkalinity was observed to be higher for the reactors yielding higher methane. The increase in alkalinity is mainly due to the activity of methanogens, which tend to generate alkaline conditions due to the formation of carbon dioxide and ammonia.

The VFA/TA ratio is usually measured as a stability index of AD, and a value less than 0.4 is preferred for a stable digestion process (Li et al., 2018). All the systems had a VFA/TA value ranging between 0.1 and 0.3 before and after anaerobic digestion, confirming good stability and buffering capacity of batch anaerobic reactors.

3.5.3. Organic matter removal efficiency of digesters

Fig. 4a and b shows the organic matter removal efficiency (OMRE) of digesters for individual and combined pretreatments in terms of sCOD, TS and VS removal. The sCOD, TS and VS removal during digestion for control was 44, 41 and 56%, respectively. Reactors fed with alkali pretreated DWAS showed maximum OMRE in case of 0.4% NaOH

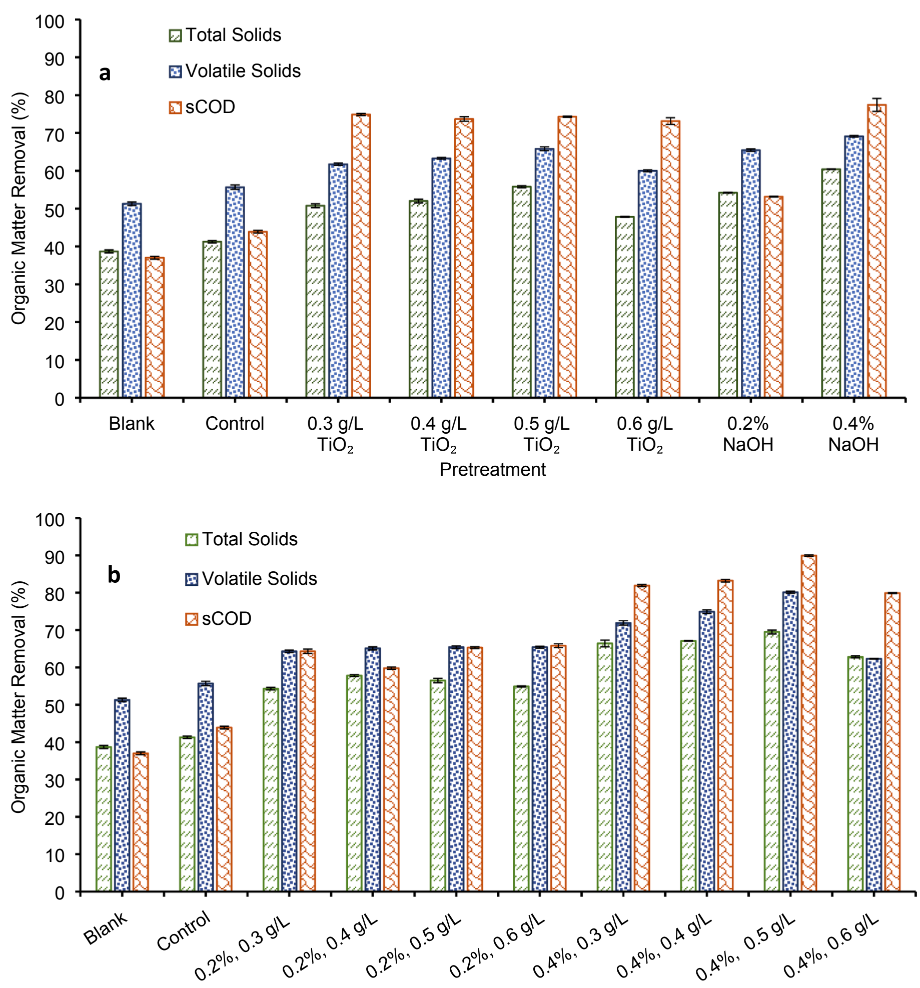


Fig. 4. Organic matter removal efficiency of digesters for NaOH and TiO_2 pretreated dewatered waste activated sludge (DWAS) (a) and NaOH–TiO_2 pretreated DWAS (b).

pretreatment. In this case, the removal of sCOD, TS and VS were 77, 60 and 69%, which was 76, 47 and 24% higher as compared to control. Lin et al. (2009) reported sCOD removal in the range of 83–93% from the pulp and paper sludge pretreated at doses of 4–16 g NaOH/100 g TS_{sludge}. Higher removal in sCOD observed by Lin et al. (2009) might be due to more delignification of biomass, making it more accessible during AD when compared with lesser sCOD removal in the present study. On the other hand, lesser removal of sCOD in the present study might be due to the use of lesser doses of NaOH per unit weight of sludge dry solids, i. e., 2–4 g NaOH/100 g TS_{sludge}.

In the case of TiO_2 NPs pretreatment, 0.5 g/L showed 74, 56 and 66% removal of sCOD, TS and VS, which was 69, 35 and 17% higher, respectively as compared to control. Anjum et al. (2018a) reported 60.6% OMRE during AD after photocatalytic pretreatment of WAS using visible light active ZnO-ZnS nanocomposite, which consequently yielded 1.6 folds higher biogas as compared to raw sludge. The reason for lesser removal efficiency in the cited study could be the use of visible light active nanoparticles, whereas in the present study, UV light active TiO_2 nanoparticles were used for pretreatment. Similarly, in combined pretreatment of the present study, 0.4% NaOH-0.5 g/L TiO_2 pretreated DWAS showed maximum OMRE of 69.5, 80.1 and 89.9% in terms of sCOD, TS and VS respectively and consequently produced 1.71 folds higher biogas as compared to raw sludge. These results are better than the findings of aforementioned study. Significant ($p < 0.05$) differences in organic matter removal from all digesters were observed as compared to control. Pretreatment enhanced OMRE of the digesters by transforming insoluble organic compounds to soluble organic compounds, leading to the release of internal organic matter of DWAS, and its conversion into methane in the subsequent process.

3.6. Methane production data validation by kinetic model

Modified Gompertz model was applied to average cumulative methane production data for control and pretreated DWAS. Kinetic parameters of methane production predicted by the model are shown in Table 3. Results show that the simulated maximum methane yield is very similar to actual cumulative methane values, which is further supported by the high values of correlation coefficient (R^2), ranging from 0.988 to 0.994. Overall, the model predicted an increase in methane production rate and decrease in lag phase time with increase in doses of alkaline, photocatalytic, and combined pretreatments. Moreover, the maximum methane production rate for alkaline-photocatalysis pretreated DWAS was higher and lag phase time was shorter than control, and individual alkaline and photocatalytic pretreatments. Combined pretreatment dose of 0.4% NaOH-0.5 g/L TiO_2 showed the shortest lag phase as compared to all other pretreatments and control. The reason might be the higher availability of soluble organic matter to the activated inoculum, which did not need much time to acclimate to a new environment and perform the activity. Rajput et al. (2018) compared the modified Gompertz model with transference and logistic function models for analyzing the effect of pretreatment on biogas production of biomass and verified that the modified Gompertz model gave the best-fitted results for biogas production. The modified Gompertz model results make it suitable to predict changes in methane production (Choi et al., 2018).

4. Conclusions

This study showed improved sludge solubilization and subsequent anaerobic digestion due to pretreatment. Alkaline-photocatalytic pretreatment showed considerably increased solubilization of organic matter and methane production as compared to alkaline and photocatalytic individual pretreatments. Methane production from DWAS at combined pretreatment dose of 0.4% NaOH-0.5 g/L TiO_2 increased significantly as compared to individual alkaline and photocatalytic pretreatments. It showed 71.1% enhanced methane yield over control, signifying higher utilization of substrate during anaerobic digestion.

Table 3
Parameters of modified Gompertz model for control and pretreated DWAS.

Pretreatment	R_{max} (N mL/g VS-d)	P_b (N mL/g VS)	λ (d)	R^2
Control	19.8	266	1.8	0.989
0.3 g/L TiO_2	22.9	308	1.7	0.993
0.4 g/L TiO_2	25.7	319	1.4	0.990
0.5 g/L TiO_2	29.9	349	1.3	0.992
0.6 g/L TiO_2	24.6	312	1.1	0.990
0.2% NaOH	30.9	360	1.4	0.992
0.4% NaOH	30.56	396	1.1	0.990
0.2% NaOH-0.3 g/L TiO_2	25.2	334	1	0.989
0.2% NaOH-0.4 g/L TiO_2	25.4	336	1.1	0.989
0.2% NaOH-0.5 g/L TiO_2	27.7	385	1	0.988
0.2% NaOH-0.6 g/L TiO_2	26.8	373	1.1	0.988
0.4% NaOH-0.3 g/L TiO_2	32.3	400	1.4	0.993
0.4% NaOH-0.4 g/L TiO_2	31.1	397	1.5	0.994
0.4% NaOH-0.5 g/L TiO_2	37.7	454	1.3	0.994
0.4% NaOH-0.6 g/L TiO_2	35.2	396	1.9	0.991

CRediT authorship contribution statement

Ayesha Maryam: Conceptualization, Methodology, Investigation, Formal analysis, Writing - original draft. **Zeshan:** Conceptualization, Supervision, Project administration, Funding acquisition. **Malik Badshah:** Data curation, Validation. **Mariam Sabeeh:** Methodology, Investigation. **Sher Jamal Khan:** Resources, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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